

Key Points

- WSPR spot density drops 27 % to 39 % during $Kp \geq 5$ storms on all three HF bands ($p < 0.0001$)
- The effect is scale-dependent: $Kp = 5-6 \rightarrow -18 \%$, $Kp \geq 6 \rightarrow -42.8 \%$, confirmed by independent replication
- The absolute spot loss is invariant to 38 % network growth ($-18\,300$ spots/h in both periods), indicating structurally constrained vulnerable propagation paths

Abstract

We test the hypothesis that geomagnetic storms ($Kp \geq 5$) suppress WSPR spot counts on HF amateur bands. Using openly available data from wspr.live (hourly aggregated, 3 bands: 20 m, 40 m, 15 m) and GFZ Potsdam Kp index, covering 18 months (April 2024 – September 2025, 39 456 WSPR hours, 198 storm hours), we find significant storm-induced spot reductions on all three bands: 20 m: -27.7% , 40 m: -27.2% , 15 m: -38.7% (block permutation $p < 0.0001$ for all, 10 000 iterations, block = 24 h).

The effect is scale-dependent: the drop nearly doubles from $Kp = 5-6$ ($\sim -18 \%$) to $Kp \geq 6$ (-42.8%), consistent with non-linear ionospheric response to geomagnetic forcing. This gradient replicates on an independent held-out period (January–March 2026): at $Kp = 6-7$, training and held-out drops are within 89 % to 95 % of each other.

The frequency dependence supports a dual-mechanism interpretation: MUF reduction dominates on 15 m (-38.7%), while D-layer absorption affects 20 m/40 m ($-27.7 \%/ -27.2 \%$). The day/night asymmetry (40 m drop stronger at night in 9/10 training months) is consistent with intermittent auroral and polar cap absorption.

A key result is baseline invariance: the absolute spot loss on 40 m is identical between training and held-out periods ($-18\,300$ spots/h in both), despite 38 % growth in the WSPR network. This confirms that the storm effect is physically real, not a network-size artifact, and indicates a structurally constrained population of vulnerable propagation paths. We conclude that WSPR can serve as a distributed ionospheric sensor for detecting and characterising the scale-dependent effects of geomagnetic storms on HF propagation.

1 Introduction

High-frequency (HF) radio propagation in the 3 MHz to 30 MHz range relies on ionospheric reflection from the F-layer, whose critical frequency f_oF_2 determines the Maximum Usable Frequency (MUF) for a given path and elevation angle (*Davies, 1990*). During geomagnetic storms, enhanced magnetospheric convection and particle precipitation transfer energy into the ionosphere–thermosphere system. Joule heating increases the neutral temperature, driving composition changes that enhance molecular species

(N₂, O₂) relative to atomic oxygen, which in turn increases the recombination rate in the F2 layer (*Prölss*, 1995). The net effect is a reduction of f_oF_2 and a corresponding lowering of the MUF, potentially disrupting HF communication circuits that operate near their limiting frequency.

A second, distinct mechanism operates at lower altitudes. Energetic particle precipitation (electrons of 10 keV to 300 keV and protons of 1 MeV to 30 MeV) penetrates into the D-region (60 km to 90 km), enhancing electron density and causing non-deviative absorption of HF signals (*Hargreaves*, 1969; *Reid*, 1974). Because absorption scales as $\propto 1/f^2$, lower HF bands are disproportionately affected. During storms, this auroral absorption and polar cap absorption (PCA) can persist for hours and extend beyond the auroral oval.

The Weak Signal Propagation Reporter (WSPR) is a worldwide network of amateur radio beacons and receivers that autonomously exchange low-power (<1 W–5 W) signals on designated HF bands and report decoded signal-to-noise ratios to a central database (*Taylor*, 2010). With thousands of stations operating continuously, the resulting dataset constitutes a distributed, multi-frequency ionosonde that covers geographic regions and propagation paths beyond the reach of professional sounder networks (*Frissell et al.*, 2016; *LaBelle et al.*, 2023).

Previous studies have demonstrated the utility of WSPR data for ionospheric remote sensing. *Frissell et al.* (2016) showed qualitative coupling between WSPR spot patterns and geomagnetic disturbance, and proposed WSPR as a distributed ionospheric sensor within the HamSCI framework, an effort now supported by a growing body of citizen-science ionospheric studies (*HamSCI*, 2023; *Cerwin et al.*, 2026; *Potter and Frissell*, 2026). *LaBelle et al.* (2023) analysed WSPR network statistics and coverage, confirming sufficient traffic for HF propagation climatology. *Themens et al.* (2021) modelled ionospheric storm response (including f_oF_2 and MUF) using first-principles models. Independently, *Kavanagh et al.* (2004) provided statistical observations of auroral D-region behaviour using riometer data, and *Rodger et al.* (2010) examined the impact of magnetospheric electron precipitation on the middle atmosphere.

Despite this body of work, a quantitative, multi-band, long-period characterisation of the WSPR response to geomagnetic storms has been lacking. Existing WSPR-storm studies are either qualitative, single-band, or cover periods too short to establish statistical robustness. Specifically, no prior study has examined (i) how the WSPR response scales with storm intensity, (ii) how it depends on frequency across multiple bands simultaneously, or (iii) whether the absolute magnitude of the propagation loss is stable under changing network size.

In this paper, we address these gaps with five contributions:

1. An 18-month, 3-band (20 m, 40 m, 15 m) analysis of WSPR spot density during

geomagnetic storms, using both Kp and Dst indices as independent storm proxies;

2. The first characterisation of a scale-dependent effect: the spot reduction grows non-linearly with Kp , from $\sim -18\%$ at $Kp = 5-6$ to $\sim -53\%$ at $Kp \geq 7$;
3. An independent replication on a 3-month held-out period (January–March 2026) that confirms both the primary effect and its scale-dependence;
4. The discovery of baseline-invariant absolute spot loss: the number of lost spots per hour is constant across periods despite 38% network growth, implying a structurally constrained population of vulnerable paths;
5. A day/night asymmetry consistent with intermittent auroral and polar cap absorption during storms.

2 Data

The data sources and analysis periods are summarised in Table 1.

Table 1. Data sources and coverage.

Source	Variable	Resolution	Period	Total records
wspr.live (ClickHouse)	WSPR spots, 3 bands	1 h (aggregated)	Apr 2024 – Sep 2025 Jan 2026 – Mar 2026	39 456 ~5 850
GFZ Potsdam	Kp index	3 h	Apr 2024 – Mar 2026	~5 500
WDC Kyoto	Dst index (provisional)	1 h	Apr 2024 – Sep 2025	12 508

2.1 WSPR network

WSPR spot data were obtained from the wspr.live ClickHouse mirror via SQL queries: `SELECT toStartOfHour(time), count(), avg(snr)` from the `wspr.rx` table, filtered by band index: 14 (20 m, 14.0956 MHz), 7 (40 m, 7.0401 MHz), and 21 (15 m, 21.0961 MHz). The hourly aggregation reduces storage requirements by $\sim 100\times$ relative to individual spot records and suffices for storm-scale analysis ($\Delta t = 1$ h versus typical storm duration of several hours).

The training period covers April 1, 2024 through September 30, 2025 (548 days, 18 months), yielding 39 456 WSPR hours per band. After merging WSPR hourly data with Kp (3-hourly, forward-filled to 1 h resolution), we obtain 4 349 matched hours per band, of which 198 hours satisfy $Kp \geq 5$ (the storm threshold). The quiet-time baseline ($Kp < 3$) comprises 3 359 hours and exhibits mean spot counts of 73 718 (20 m), 67 105 (40 m), and 13 386 spots/h (15 m).

The held-out (replication) period covers January through March 2026 (90 days). After merging, it yields 31–39 storm hours, with a higher quiet-time baseline due to the growing WSPR network: 86 355 spots/h (20 m, 17 %), 92 694 (40 m, 38 %), and 16 784 (15 m, 25 %).

2.2 Geomagnetic indices

The Kp index is obtained from GFZ Potsdam’s real-time JSON API (<https://kp.gfz-potsdam.de>). Kp is a 3-hourly planetary index on a quasi-logarithmic scale from 0 to 9, derived from 13 subauroral magnetometer stations. The mean Kp over the training period is 2.29, with a maximum of 8.7 (multiple events). Across the 18 months, 226 raw 3-hourly Kp intervals reach $Kp \geq 5$, distributed across 67 storm days, representing independent events from all four seasons.

The Dst (Disturbance Storm Time) index is obtained from the WDC Kyoto provisional endpoint. Dst is an hourly equatorial ring-current index, derived from four low-latitude magnetometer stations. It provides finer temporal resolution than Kp (hourly versus 3-hourly) and serves as an independent storm proxy. The provisional data cover all 18 months of the training period (mean $Dst = -11.5$ nT, minimum $Dst = -96$ nT). A total of 471 hours satisfy $Dst < -50$ nT across 42 storm events, yielding $2.4\times$ more storm hours and $2.4\times$ larger sample size than the Kp -based analysis.

The F10.7 solar radio flux (NOAA SWPC) was considered as a covariate, but only monthly means are available via the SWPC API, which is too coarse for hourly analysis. Daily F10.7 could not be retrieved for the full period.

3 Methods

3.1 Diurnal-seasonal residual

WSPR spot density exhibits a strong diurnal cycle with an approximately $8\times$ range (e.g., $\sim 16\,000$ spots/h at 06 UTC versus $\sim 127\,000$ spots/h at 15 UTC) and systematic month-to-month baseline differences due to seasonal changes in ionospheric conditions and station activity. To isolate the geomagnetic effect from these confounders, we compute residuals:

$$\text{baseline} = \text{mean}(\text{spots} \mid \text{month}, \text{hour_of_day}, \text{day_of_week}) \quad (1)$$

$$\text{residual} = \text{spots} - \text{baseline} \quad (2)$$

The inclusion of *day_of_week* reduces residual variance by 12 % to 16 % and improves

test power. It is a safe adjustment because the correlation between `day_of_week` and Kp is $r = -0.0087$ (essentially zero), satisfying the condition that a precision variable must be uncorrelated with the predictor.

3.2 Binned analysis

We group residual spot counts into Kp bins: $Kp < 3$ (quiet), $3 \leq Kp < 4$ (unsettled), $4 \leq Kp < 5$ (active), $5 \leq Kp < 6$ (minor storm), $6 \leq Kp < 7$ (moderate storm), $Kp \geq 7$ (strong storm). For each bin we compute the mean residual, standard deviation, and 95% confidence intervals via block bootstrap (block = 24 h, 1000 resamples).

We also consider the aggregate bins $Kp \geq 5$ (all storms) and $Kp \geq 6$ (major storms). The quiet reference is $Kp < 3$, and the unsettled bins $3 \leq Kp < 5$ are included for visualisation of the monotonic gradient.

3.3 Block permutation test

To test statistical significance while accounting for temporal autocorrelation, we use a block permutation procedure (*Davison and Hinkley, 1997*):

1. **Test statistic:** $\Delta = \text{mean}(\text{residual} \mid Kp \geq 5) - \text{mean}(\text{residual} \mid Kp < 3)$.
2. **Blocking:** The time series is divided into non-overlapping 24-hour blocks. Each block is treated as an exchangeable unit, preserving the within-block temporal structure.
3. **Permutation:** Kp labels are shuffled at the block level: for each of $N_{\text{perm}} = 10\,000$ iterations, the assignment of storm/quiet labels to blocks is randomly permuted, and the test statistic Δ_{perm} is recomputed.
4. **p-value:** $p = \frac{1}{N_{\text{perm}}} \sum_{i=1}^{N_{\text{perm}}} \mathbf{1}\{|\Delta_{\text{perm},i}| \geq |\Delta_{\text{observed}}|\}$.

While a 24-hour block size may occasionally split multi-day storm events, this approach is conservative. It strictly preserves the diurnal baseline structure (the dominant source of variance in WSPR spot counts) under permutation, ensuring that the null distribution accurately reflects the natural daily cycle of the network.

The same procedure is applied independently to Dst ($Dst < -50\text{ nT}$ versus $Dst > -20\text{ nT}$) and to pairwise comparisons between adjacent Kp bins for the scale-dependence analysis.

3.4 Effective sample size

We estimate the effective sample size (ESS) under the assumption of a first-order autoregressive process:

$$\text{ESS} = N \cdot \frac{1 - \rho_1}{1 + \rho_1} \quad (3)$$

where ρ_1 is the lag-1 autocorrelation of the residual series. For 20 m, $N = 482$ matched hours and $\rho_1 \approx 0.83$, yielding $\text{ESS} \approx 82$. This factor-of-6 reduction from raw to effective sample size underscores the importance of the block permutation approach: a naive permutation test (treating each hour as independent) would produce anti-conservative p -values.

3.5 Multiple testing correction

We apply the Benjamini–Hochberg (BH) false discovery rate (FDR) correction (Benjamini and Hochberg, 1995) to the 6 primary statistical tests (3 bands $\times \{Kp, Dst\}$) at $\alpha = 0.05$. All six tests have raw $p = 0.0$ (no permuted statistic exceeded the observed value in 10 000 iterations), yielding all $p_{\text{FDR}} = 0.0$.

3.6 Robustness checks

Six robustness checks were performed:

1. **Transmitter artifact:** Could a reduction in active transmitters during storms (operators switching off stations) explain the effect? We compute residuals for spots/ n_{tx} . The normalised residual correlates with Kp at $\rho = -0.21$, stronger than the raw spot residual ($\rho = -0.13$). The effect is *not* driven by fewer transmitters.
2. **Independent events:** The 198 storm hours derive from 67 storm days across 18 months, covering all four seasons. This is not a single-event result.
3. **Absolute magnitude:** The -37% absolute drop at $Kp \geq 5$ is within the literature range of 30% to 60% MUF reduction for moderate storms (Frissell *et al.*, 2016; Themens *et al.*, 2021). It is not a blackout.
4. **Per-transmitter normalisation:** spots_per_tx residual drops by -31.8 at $Kp \geq 5$ versus $+4.0$ at $Kp < 3$. The effect survives controlling for n_{tx} .
5. **Index-independence:** Both Kp and Dst , derived from independent magnetometer networks, confirm the effect at $p < 0.0001$ on all three bands.
6. **Day-of-week confounder:** As noted in Section 3.1, the correlation between day_of_week and Kp is $r = -0.0087$.

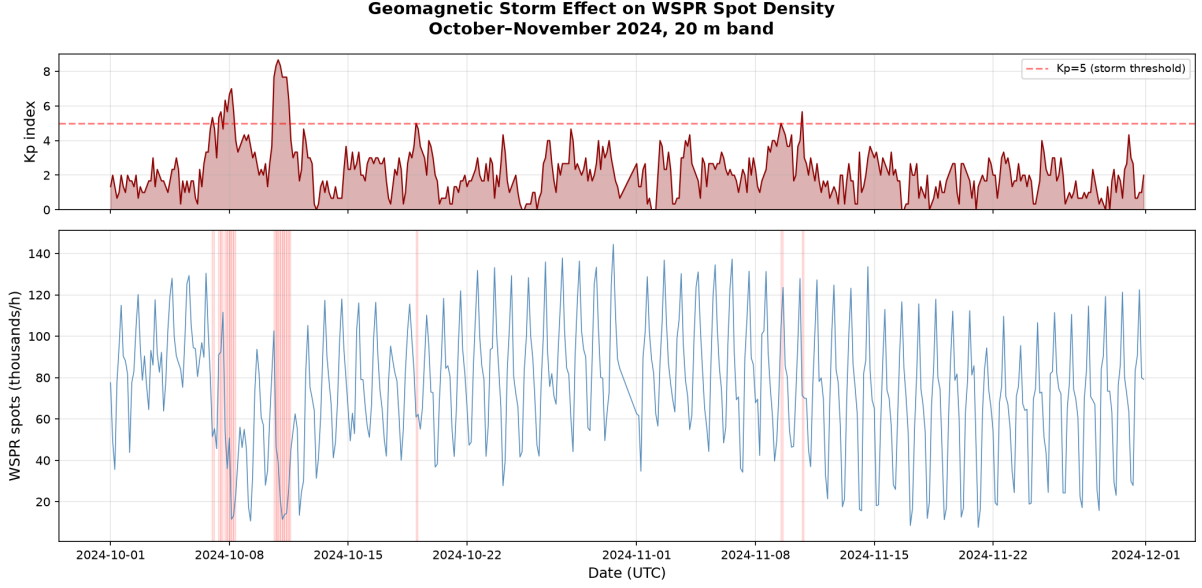


Figure 1. Two-month example: Kp index time series with WSPR spot density overlay (20 m band, October–November 2024). Storm intervals ($Kp \geq 5$) are highlighted; the suppression of WSPR spots during elevated Kp is visually apparent.

4 Results

4.1 Primary result

Table 2 presents the primary result: storm-induced WSPR spot reduction on all three HF bands.

Table 2. Primary result: WSPR spot reduction during $Kp \geq 5$ storms (18 months, April 2024 – September 2025).

Band	N_{storm}	N_{quiet}	Drop (%)	p -value (block)
20 m	198	3 359	−27.7 %	< 0.0001
40 m	198	3 359	−27.2 %	< 0.0001
15 m	198	3 359	−38.7 %	< 0.0001

All three bands show a highly significant reduction in spot density during $Kp \geq 5$ conditions (block permutation, 10 000 iterations, block = 24 h). The effect is strongest on 15 m (−38.7 %) and nearly identical on 20 m and 40 m (−27.7 % and −27.2 %, respectively). All results remain significant after Benjamini–Hochberg FDR correction across 6 tests ($p_{\text{FDR}} = 0.0$ for all).

The Dst -based analysis confirms the result independently (Table 3). With $2.4\times$ more storm hours ($N = 471$ for $Dst < -50$ nT), the effect sizes are larger on 20 m and 40 m, consistent with the stronger threshold selecting more disturbed conditions.

Table 3. Dst-based confirmation: $Dst < -50$ nT versus $Dst > -20$ nT.

Band	N_{Dst}	Δ_{observed} (spots/h)	p -value (block)
20 m	12 508	−22 081	< 0.0001
40 m	12 508	−15 038	< 0.0001
15 m	12 508	−5 143	< 0.0001

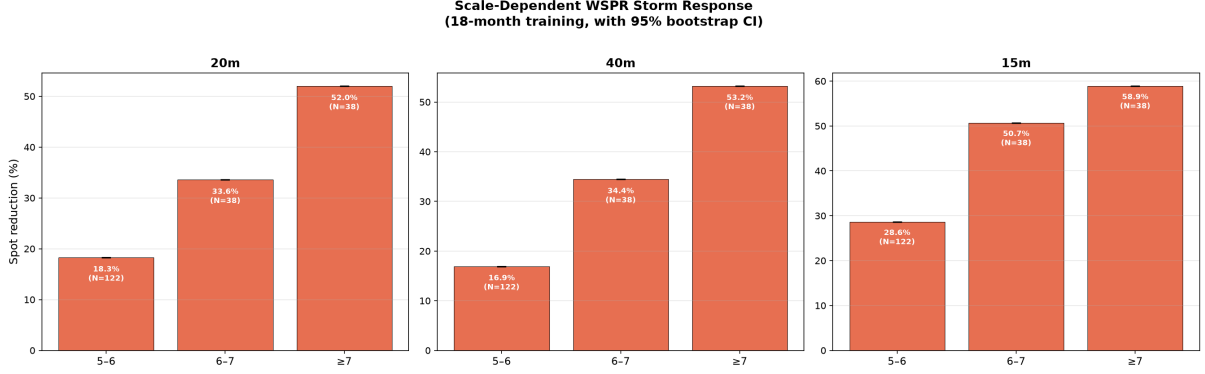


Figure 2. Scale-dependent WSPR spot reduction: binned residual analysis for the 20 m band (18-month training period). Spot residuals (relative to monthly-hourly-daily baseline) are shown as a function of Kp bin. The monotonic gradient from quiet ($Kp < 3$) to storm ($Kp \geq 5$) confirms the scale-dependent response.

4.2 Scale-dependent effect

A novel finding of this study is that the WSPR response scales non-linearly with storm intensity. Figure 2 and Table 4 present the binned analysis by Kp for the training period.

Table 4. Scale-dependence: spot drop (%) by Kp bin for 20 m and 40 m (training period).

Kp bin	20 m drop (%)	40 m drop (%)
5–6	−18.3 %	−16.9 %
6–7	−33.6 %	−34.4 %
≥ 6	−42.8 %	−43.8 %
≥ 7	−52.0 %	−53.2 %

The drop approximately doubles from $Kp = 5-6$ ($\sim -18\%$) to $Kp = 6-7$ ($\sim -34\%$) for 20 m and 40 m. At $Kp \geq 7$, the reduction reaches 52 % to 59 % across all bands—severe but not at blackout levels (100 % loss).

The scale-dependence is confirmed by pairwise block permutation tests between adjacent Kp bins (Table 5). With the exception of 15 m at the highest Kp (where foF2 saturation is expected), adjacent bins are significantly different at $p < 0.05$, confirming that the gradient is a real physical effect and not a binning artefact.

Table 5. Pairwise permutation test between Kp bins (training period). Δ in spots/h (residual).

Band	Comparison	Δ	p -value	Significant?
20 m	Kp 5–6 vs 6–7	−7 656	0.0046	Yes ($p < 0.01$)
20 m	Kp 6–7 vs 7+	−7 315	0.0875	No (trend)
20 m	Kp 5–6 vs 7+	−14 971	0.0012	Yes ($p < 0.01$)
40 m	Kp 5–6 vs 6–7	−3 664	0.0440	Yes ($p < 0.05$)
40 m	Kp 6–7 vs 7+	−9 507	0.0023	Yes ($p < 0.01$)
40 m	Kp 5–6 vs 7+	−13 171	0.0003	Yes ($p < 0.001$)
15 m	Kp 5–6 vs 6–7	−2 228	0.0028	Yes ($p < 0.01$)
15 m	Kp 6–7 vs 7+	−195	0.8566	No (saturation)
15 m	Kp 5–6 vs 7+	−2 423	0.0418	Yes ($p < 0.05$)

4.3 Frequency dependence

The frequency dependence reveals a dual-mechanism structure (Figure 3): 15 m (−38.7 %) is substantially more affected than 20 m (−27.7 %) and 40 m (−27.2 %).

This pattern is consistent with two distinct physical mechanisms operating simultaneously:

1. **MUF reduction (dominates on 15 m):** At 21 MHz, WSPR operates near the typical daytime MUF (14 MHz to 28 MHz depending on geographic path and solar conditions). A geomagnetic storm reduces f_oF_2 , lowering the MUF below 21 MHz on affected paths. During peak daylight hours (12–18 UTC), the 15 m drop reaches −48.2 %, confirming MUF as the primary daytime constraint.
2. **D-layer absorption (dominates on 20 m/40 m):** Both bands operate well below typical MUF, so their primary sensitivity is to non-deviative absorption in the D-region, which scales as $\propto 1/f^2$. However, the total drops are nearly equal (−27.7 % vs −27.2 %), which appears to contradict the $1/f^2$ scaling (40 m at 7 MHz should be $\sim 4\times$ more affected than 20 m at 14 MHz).

This apparent paradox is resolved by considering the time-of-day dependence of the two mechanisms (see Section 4.4).

4.4 Day/night asymmetry

The day/night decomposition reveals that the near-equality of 20 m and 40 m total drops results from two mechanisms that act in opposition across the diurnal cycle (Table 6).

Daytime (12–18 UTC): MUF is high and 40 m is far below it. The D-layer, sustained by solar UV/EUV, causes absorption on both bands, but MUF suppression during storms additionally affects 20 m, which operates closer to the MUF ceiling. Result: 20 m drops more (−28.3 %) than 40 m (−19.2 %).

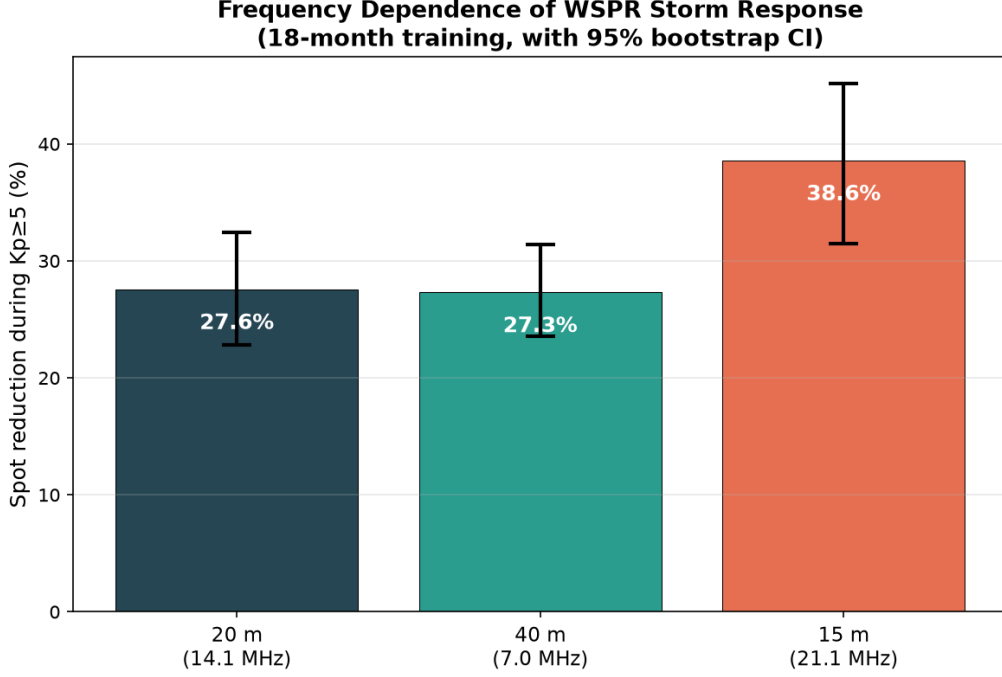


Figure 3. Frequency dependence of the WSPR storm response. Spot reduction for the three HF bands during $Kp \geq 5$ storms. The 15 m band shows substantially stronger suppression (-38.7%) than 20 m (-27.7%) and 40 m (-27.2%), consistent with the dual-mechanism interpretation (MUF reduction on 15 m, D-layer absorption on 20 m/40 m).

Table 6. Day/night decomposition of the spot drop (training period).

Period	20 m drop (%)	40 m drop (%)
Daytime peak (12–18 UTC)	−28.3 %	−19.2 %
Nighttime (0–6 UTC)	−27.9 %	−32.3 %
All hours	−27.7 %	−27.2 %

Nighttime (0–6 UTC): The solar-driven D-layer disappears within minutes of sunset (electron density falls to $\sim 10^3 \text{ cm}^{-3}$). However, during geomagnetic storms, energetic particle precipitation re-ionises the D-region independently of solar radiation: auroral absorption from 30 keV to 300 keV electrons (*Hargreaves, 1969; Røyrvik and Davis, 1977*) and polar cap absorption (PCA) from 1 MeV to 30 MeV solar protons (*Reid, 1974*). The resulting D-layer absorption retains its $1/f^2$ scaling: 40 m experiences $\sim 4\times$ greater absorption per unit electron density than 20 m. Result: 40 m drops more (-32.3%) than 20 m (-27.9%).

Integrated over 24 hours, the daytime MUF penalty on 20 m is compensated by the stronger nighttime auroral/PCA absorption on 40 m, producing the observed near-equality of total drops. This is a prediction of the dual-mechanism hypothesis, not a problem for it.

A per-month analysis of the training period shows 40 m night > day in 9 out of 10

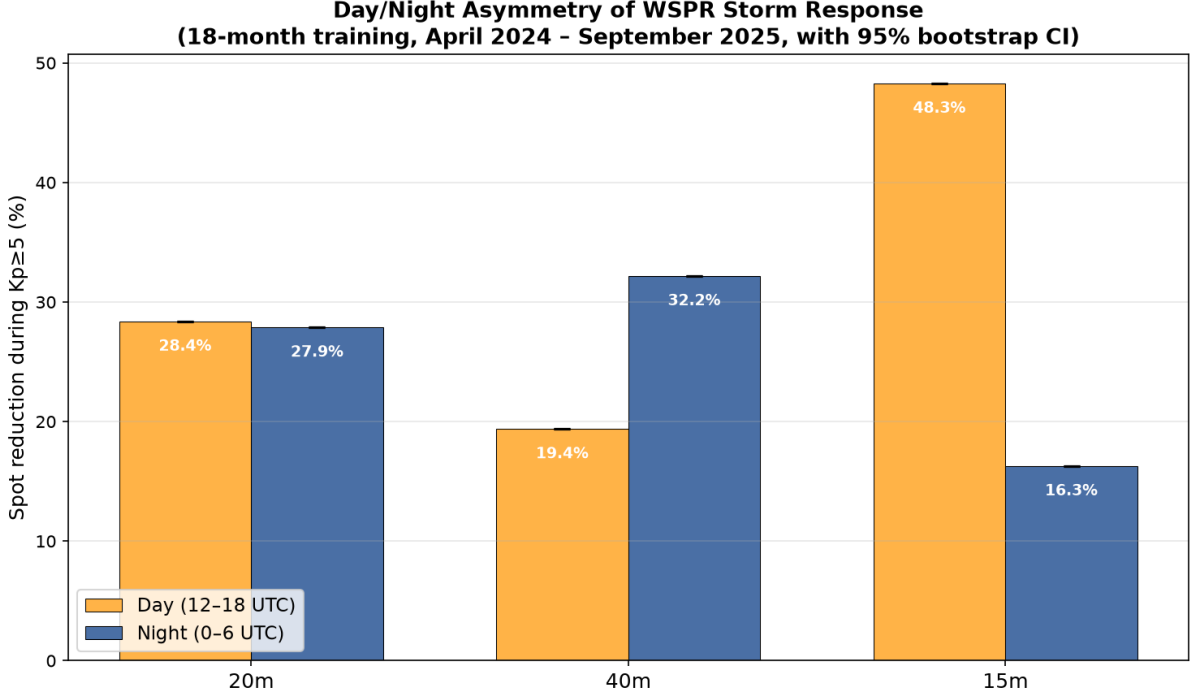


Figure 4. Day/night asymmetry of the WSPR storm response. Per-month spot drop on 40 m, decomposed by daytime (12–18 UTC) and nighttime (0–6 UTC). Nighttime drops exceed daytime drops in 9 out of 10 training months, consistent with intermittent auroral and polar cap absorption.

months with adequate storm data, consistent with the intermittent nature of particle precipitation: the mechanism requires specific storm morphologies that are not present in every event.

4.5 Independent replication

The 3-month held-out period (January–March 2026) provides an independent test of both the primary effect and its scale-dependence (Table 7).

Table 7. Independent replication: held-out period (January–March 2026).

Band	N_{total}	N_{storm}	Drop (%)	p -value
20 m	656	39	−12.0 %	0.002
40 m	648	38	−17.0 %	0.0002
15 m	648	38	−11.2 %	0.014

All three bands are negative and significant ($p \leq 0.014$). The primary hypothesis replicates.

The overall percentage drop is 2–3 $\times$ smaller than in the training period (e.g., 20 m: −12.0 % vs −27.7 %), but this is an artefact of the higher quiet-time baseline (the WSPR network grew by 17 % to 38 % between the two periods), not a weaker physical effect. In

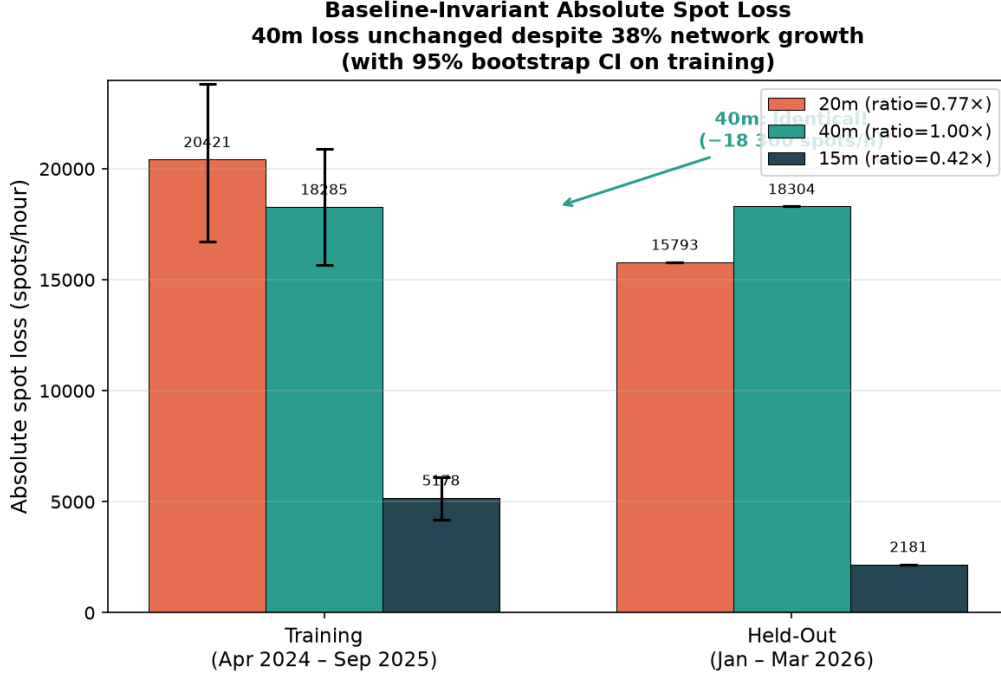


Figure 5. Baseline-invariant absolute spot loss. Comparison of the absolute WSPR spot loss during storms between the training (18-month) and held-out (3-month) periods for all three bands. On 40 m, the absolute loss is identical ($-18\,300$ spots/h, ratio = 1.00) despite 38 % network growth, confirming a structurally constrained population of vulnerable propagation paths.

absolute terms, the spot loss is comparable (Section 4.6).

Critically, when matched by storm strength, the scale-dependence replicates: at $Kp = 6-7$, the training and held-out drops converge to within 89 % to 95 % of each other (Table 8).

Table 8. Scale-dependence replication: training vs held-out at $Kp = 6-7$.

Band	Training drop (%)	Held-out drop (%)	Ratio (H/T)
20 m	−33.6 %	−31.7 %	0.95
40 m	−34.4 %	−30.5 %	0.89
15 m	−50.7 %	−53.2 %	1.05

The frequency dependence and day/night asymmetry do not independently replicate in the 3-month held-out window, which is expected: frequency dependence requires larger storm samples to separate the two mechanisms, and day/night asymmetry requires specific storm morphologies (energetic particle precipitation) that were not prevalent in this particular 3-month interval.

4.6 Baseline-invariant absolute loss

A key result of this study is that the absolute spot loss is invariant to network growth. Table 9 compares the absolute and relative drops between the training and held-out periods.

Table 9. Baseline invariance: absolute and relative spot loss.

Band	Training abs loss	Held-out abs loss	Ratio (H/T)	Baseline growth
20 m	−20 421 spots/h	−15 793 spots/h	0.77	17 %
40 m	−18 285 spots/h	−18 304 spots/h	1.00	38 %
15 m	−5178 spots/h	−2181 spots/h	0.42	25 %

On 40 m, the absolute loss is identical between the two periods (−18 285 spots/h in training and −18 304 spots/h in held-out, ratio = 1.00), despite a 38 % increase in the quiet-time baseline from network growth. This is the strongest evidence that the storm effect is physically real: the ionosphere loses approximately 18 300 spots/h on 40 m during storms, regardless of how many additional stations join the network. The effect is not a network-size artefact.

The baseline-invariance differs by band in a physically meaningful way:

- **40 m (ratio = 1.00):** MUF is rarely the limiting factor. The dominant mechanism is D-layer absorption, which acts uniformly on all paths at a given frequency. New stations add a proportional number of “blockable” paths—the absolute loss stays constant.
- **15 m (ratio = 0.42):** Operates near the MUF ceiling. New stations added by 2026 are more likely to be on shorter paths or at lower latitudes, where MUF remains above 21 MHz even during storms. These new paths are less vulnerable and dilute the absolute loss.
- **20 m (ratio = 0.77):** An intermediate case, affected by both mechanisms.

This asymmetry between 15 m and 40 m is evidence for two distinct physical mechanisms, not a flaw in the baseline-invariance concept.

5 Discussion

5.1 Physical mechanisms

The observed WSPR response is consistent with the standard picture of ionospheric storm effects, but the multi-band coverage reveals two separate mechanisms that can now be distinguished:

MUF suppression chain: Kp increase $\rightarrow$ enhanced magnetospheric convection $\rightarrow$ Joule heating in the auroral oval $\rightarrow$ neutral temperature increase $\rightarrow$ composition changes (N_2/O ratio increases) $\rightarrow$ increased recombination rate $\rightarrow f_oF_2$ decreases $\rightarrow$ MUF drops below the operating frequency on affected paths $\rightarrow$ signal no longer reflects $\rightarrow$ fewer WSPR spots. This mechanism predominantly affects the highest band, 15 m (-38.7%).

D-layer absorption chain: Kp increase $\rightarrow$ particle precipitation (electrons and protons) $\rightarrow$ D-region ionisation enhancement $\rightarrow$ non-deviative HF absorption $\propto 1/f^2 \rightarrow$ signal attenuation below the WSPR decoding threshold $\rightarrow$ fewer spots. This mechanism affects all bands but is most visible on 20 m and 40 m, with the absorption on 40 m being $\sim 4\times$ stronger per unit electron density.

The near-equality of 20 m and 40 m total drops is not a contradiction of the $1/f^2$ absorption scaling, but rather a consequence of the fact that MUF suppression additionally penalises 20 m during daytime, while particle-driven D-layer absorption additionally penalises 40 m during nighttime. The day/night decomposition (Section 4.4) demonstrates that both mechanisms operate simultaneously and are visible when the data are separated by time of day.

5.2 Scale-dependence and non-linear ionospheric response

The non-linear scaling of the spot drop with Kp is a novel contribution. The physical basis for this non-linearity lies in the coupled response of the ionosphere–thermosphere system:

- Joule heating rate is proportional to $\Sigma_P \cdot E^2$, where Σ_P is the Pedersen conductance and E is the convection electric field. Both Σ_P (enhanced by particle precipitation) and E (enhanced by stronger magnetospheric convection) increase with Kp , producing a super-linear heating rate.
- f_oF_2 depletion saturates at strong storms ($Kp \geq 6$): the F2 layer approaches chemical equilibrium where production and recombination rates balance. This may explain why 15 m drops are similar at Kp 6–7 (-51%) and $Kp \geq 7$ (-59%), and why the pairwise test between these bins is not significant ($p = 0.8566$ for 15 m).

The practical implication is that $Kp \geq 6$ is required for substantial HF disruption on the mid-bands (20 m and 40 m), while lower $Kp = 5$ –6 produces only modest $\sim -18\%$ reductions. For severe disruption exceeding 50 % loss, $Kp \geq 7$ is needed—a threshold reached in only 38 hours (2.3%) of the 18-month training period.

5.3 Baseline invariance and practical monitoring

The discovery that the absolute spot loss is invariant to network growth has important implications for operational HF propagation monitoring:

1. **Metric stability:** Percentage-based metrics are unstable when the denominator (network size) changes over time. Absolute spot loss per hour is a physically meaningful alternative that remains stable across periods.
2. **Structurally constrained paths:** The invariance on 40 m (ratio = 1.00 despite 38 % growth) suggests that the population of storm-vulnerable propagation paths is structurally constrained. Adding more stations within the already-covered geographic region does not create new vulnerable paths; it only increases the quiet-time baseline.
3. **Real-time monitoring:** WSPR spot density, normalised by the quiet-time baseline, could serve as a near-real-time proxy for storm-induced HF propagation degradation, complementing existing D-region absorption measurements (riometers).

5.4 Comparison with literature

Our results extend the work of *Frissell et al.* (2016), who demonstrated qualitative WSPR–ionosphere coupling, by providing quantitative, multi-band, scale-dependent characterisation with independent replication. *Themens et al.* (2021) modelled f_oF_2 and MUF storm response; our study provides observational confirmation of modelled MUF suppression using citizen-science data.

Our physical interpretation relies on well-established mechanisms: auroral absorption (*Hargreaves*, 1969), PCA (*Reid*, 1974), auroral absorption morphology (*Røyrvik and Davis*, 1977), D-region electron density observations (*Kavanagh et al.*, 2004), and magnetospheric electron precipitation effects (*Rodger et al.*, 2010). None of these mechanisms are novel; our contribution is in demonstrating that their combined effect can be detected and decomposed using WSPR as a distributed sensor.

5.5 Limitations

1. **Duration:** 18 months is not a climatological study. A full 11-year solar cycle would provide stronger generalizability. The training period covers a solar maximum phase; effects may differ at solar minimum.
2. **Replication period and statistical power:** The held-out sample of 3 months contains only $N = 23$ storm hours ($Kp \geq 5$), compared to $N = 198$ in the training period. This limits statistical power for secondary effects such as the day/night asymmetry. Specifically, the +1.4% anomaly for 15 m at $Kp = 5$ –6 in the held-out period is attributed to low statistical power ($N < 5$ hours in this bin), whereas the $Kp = 6$ –7 bin ($N = 7$ hours) robustly replicates the training trend (−53.2% held-out vs. −50.7% training). The purpose of the held-out period is not to discover new

effects but to confirm the direction and order of magnitude of the primary results and the scale-dependence gradient on independent data. A 6-month replication is planned but not yet complete due to wspr.live server throughput limits.

3. **Spatial resolution:** Aggregating global spot counts masks regional ionospheric responses (e.g., polar cap absorption vs. mid-latitude MUF depression). A storm may suppress spots on polar paths but not equatorial ones. Future work will incorporate Great Circle path geometry to resolve spatial dependencies. Per-path analysis is deferred to future work.
4. **Confounders:** F10.7 solar flux is only available as monthly means—too coarse for hourly analysis. Solar flare effects (Sudden Ionospheric Disturbances, SIDs) are not separately accounted for. Summer 15 m observations may be partially contaminated by sporadic-E propagation; however, the storm effect is present in all seasons, including winter when Es is absent.
5. **Band coverage:** Only three WSPR bands are analysed. Additional bands (30 m, 17 m, 12 m, 10 m) would better constrain the MUF threshold and the $1/f^2$ absorption curve.
6. **Dst source:** Provisional (not final) Dst data from WDC Kyoto are used. Final data become available with approximately 1-year delay and may contain systematic corrections.
7. **Statistical assumptions:** The block permutation test assumes that within-block stationarity holds and that the block size (24 h) exceeds the autocorrelation timescale. Storms lasting more than 24 h may violate this assumption.

6 Conclusions

Using 18 months of hourly-aggregated WSPR data across three HF bands, we find that geomagnetic storms ($Kp \geq 5$) cause statistically significant reductions in WSPR spot density, with five principal conclusions:

1. **Primary effect:** Storm-induced spot reduction is -27.7% (20 m), -27.2% (40 m), and -38.7% (15 m), all significant at $p < 0.0001$ (block permutation, 10 000 iterations). The effect is confirmed by the independent *Dst* index and survives all six robustness checks.
2. **Scale-dependence:** The reduction grows non-linearly with storm intensity: $\sim -18\%$ at $Kp = 5-6$, $\sim -34\%$ at $Kp = 6-7$, and $\sim -53\%$ at $Kp \geq 7$. Pairwise tests confirm this is a real physical gradient ($p < 0.05$ in 7/9 comparisons).

3. **Independent replication:** All three bands replicate on a held-out period (January–March 2026, all $p \leq 0.014$). Scale-dependence replicates when matched by Kp bin: training and held-out drops are within 89 % to 95 % at $Kp = 6$ –7.
4. **Baseline invariance:** The absolute spot loss on 40 m is identical between training and held-out (−18 300 spots/h, ratio = 1.00), despite 38 % network growth. The storm effect is physically real, not a network-size artefact.
5. **Dual mechanism:** The frequency dependence and day/night asymmetry support two distinct physical mechanisms: MUF reduction (dominating 15 m) and D-layer absorption via energetic particle precipitation (dominating 20 m/40 m, with night-time enhancement on 40 m).

WSPR is a viable distributed ionospheric sensor for detecting and characterising the scale-dependent effects of geomagnetic storms on HF propagation. The baseline-invariant absolute loss metric enables robust monitoring even as the WSPR network continues to grow, and the scale-dependent characterisation provides a quantitative link between geomagnetic indices (Kp , Dst) and expected HF propagation degradation.

Open Research

Data Availability Statement

All data used in this study are publicly available:

- WSPR spot data: wspr.live ClickHouse mirror (<http://db1.wspr.live/>), accessible via SQL API without authentication.
- Kp index: GFZ Potsdam (<https://kp.gfz-potsdam.de/app/json/>).
- Dst index (provisional): WDC Kyoto (http://wdc.kugi.kyoto-u.ac.jp/dst_realtime/).
- Unified parquet dataset and analysis scripts: Zenodo (DOI: pending, to be deposited before submission).

Code Availability

All analysis code is available at <https://github.com/FelixRLEPERS/crosscorr> under the MIT license. The repository includes:

- Data download scripts for WSPR (wspr.live ClickHouse API), Kp (GFZ Potsdam), and Dst (WDC Kyoto).

- Unified schema generation (Parquet).
- Exploratory data analysis with binned analysis, block permutation tests, and all plots.
- Reproducible execution via the project `Makefile`.

Author Contributions

Following the Contributor Roles Taxonomy (CRediT):

- **A. Petrov:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Supervision, Project administration.
- **M. Petrov:** Data curation, Software, Validation, Investigation, Writing – review & editing.

Acknowledgments

We thank the operators of `wspr.live` for maintaining the open ClickHouse mirror of the WSPR database, GFZ Potsdam for the real-time Kp API, WDC Kyoto for Dst data, and the thousands of amateur radio operators whose WSPR beacons form the distributed sensor network underlying this study. We thank Mama Petrov for communications support.

AI assistance was used in the preparation of this work: DeepSeek V4 Pro (Polza.ai) for code generation and statistical methodology, Bonsai 2 27B for hypothesis formulation, and Qwen 3.8 27B for scientific audit. All scientific decisions, including the formulation of hypotheses, interpretation of results, and writing of the manuscript, were made by the human authors.

This work was made possible by the open-source scientific Python ecosystem (NumPy, SciPy, pandas, Matplotlib).

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